SKYLITE / SKYNET Architecture Document

(Diagram-Heavy, Maximum Detail, Word-Formatted)

1. High-Level System Architecture

1.1 Overall System Flow

```
+---------------------------+
|        Web Browser        |
|     (JSCore + HTML)       |
+-------------+-------------+
              |
              | HTTP GET / POST
              v
+-------------+-------------+
|     Web Server Layer      |
|  IIS (SkyLite) / Kestrel  |
|     (SkyNet / ASP.NET)    |
+-------------+-------------+
              |
              v
+---------------------------+
|     Framework Engine      |
|  SkyLiteHandler / IHandler|
+---------------------------+
              |
              v
+---------------------------+
|       Page Resolver       |
| Reflection-Based Routing  |
+---------------------------+
              |
              v
+---------------------------+
|        WebPage Class      |
|  Lifecycle + Rendering    |
+---------------------------+
```

```
              |
              v
+---------------------------+
|   ApiResponse Generator   |
+---------------------------+
              |
              v
+---------------------------+
|   JSCore Command Engine   |
+---------------------------+
```

2. Request Lifecycle Architecture

2.1 GET Request Pipeline

```
Browser
   |
   | GET /PageName
   v
+--------------------------+
|   Framework Middleware   |
+--------------------------+
   |
   | Resolve Page Class
   v
+--------------------------+
|      WebPage Instance    |
+--------------------------+
   |
   | OnLoad()
   v
+--------------------------+
| Load Template (HTML/TXT) |
+--------------------------+
   |
   | OnInitialized()
   v
+--------------------------+
| Modify HtmlDocument      |
+--------------------------+
```

```
   |
   | OnBeforeRender()
   v
+--------------------------+
| Inject Scripts + Translate|
+--------------------------+
   |
   | OnAfterRender()
   v
+--------------------------+
| Send Final HTML to Client |
+--------------------------+
```

2.2 POST Request Pipeline (ApiRequest)

```
Browser (JSCore)
   |
   | POST ApiRequest
   v
+-------------------------+
|   Framework Middleware    |
+-------------------------+
   |
   | Resolve Page Class
   v
+-------------------------+
|      WebPage Instance     |
+-------------------------+
   |
   | Invoke Method()
   v
+-------------------------+
|     ApiResponse Object    |
+-------------------------+
   |
   | Translate Response
   v
+-------------------------+
|   Send JSON to Browser    |
+-------------------------+
   |
   | JSCore Executes Commands
   v
```

```
   |
   | JSCore Executes Commands
   v
+-------------------------+
|   DOM Updated Dynamically |
+-------------------------+
```

3. SKYNET Middleware Architecture

3.1 ASP.NET Core Pipeline Integration

```
+-----------------------------------------+
|           ASP.NET Core Pipeline         |
+-----------------------------------------+
|  UseRouting()                           |
|  UseStaticFiles()                       |
|  UseMiddleware<IHandler>()  <-----------+
|  UseStaticHttpCurrent()                 |
|  UseEndpoints()                         |
+-----------------------------------------+
```

IHandler Responsibilities

```
+--------------------------+
|         IHandler         |
+--------------------------+
| - Intercept Request      |
| - Detect GET/POST        |
| - Resolve Page Class     |
| - Execute Lifecycle      |
| - Serialize ApiResponse  |
+--------------------------+
```

4. Page Resolution Architecture

4.1 Reflection-Based Routing

```
URL: /HR_UserProfile
        |
        v
+-------------------------+
|   Page Resolver Engine    |
+-------------------------+
| Assembly.GetType(name)   |
| Activator.CreateInstance |
+-------------------------+
        |
        v
+-------------------------+
|     HR_UserProfile        |
|        : WebPage          |
+-------------------------+
```

Naming Rule Diagram

```
Valid:
+-------------------------+
| HR_UserProfile          |
| CRM_UserProfile         |
| INV_ItemList            |
+-------------------------+

Invalid:
+-------------------------+
| UserProfile             |
| UserProfile             |  <-- Duplicate class names
+-------------------------+
```

5. WebPage Lifecycle Architecture

```
+-------------------------+
|        WebPage          |
+-------------------------+
        |
        | OnLoad()
        v
+-------------------------+
| Load Template           |
+-------------------------+
        |
        | OnInitialized()
        v
+-------------------------+
| Modify HtmlDocument     |
+-------------------------+
        |
        | OnBeforeRender()
        v
+-------------------------+
| Inject Scripts          |
| Apply Translation       |
+-------------------------+
        |
        | OnAfterRender()
        v
+-------------------------+
| Final HTML Output       |
+-------------------------+
```

6. HtmlDocument Architecture

```
+--------------------------+
|       HtmlDocument       |
+--------------------------+
| HtmlBodyText             |
| HtmlBodyAddOn            |
| HeadContent              |
| ScriptInjection          |
+--------------------------+
```

Template Loading Flow

```
Check for File:
   |
   |-- UserProfile.html
   |-- UserProfile.txt
   |
   v
Load into HtmlBodyText
```

7. ApiRequest / ApiResponse Architecture

7.1 ApiRequest Structure

```
+--------------------------+
|       ApiRequest         |
+--------------------------+
| PageName                 |
| MethodName               |
| Parameters               |
+--------------------------+
```

7.2 ApiResponse Structure

```
+-------------------------+
|       ApiResponse       |
+-------------------------+
| Commands[]              |
|  - SetValue             |
|  - UpdateHtml           |
|  - ShowDialog           |
|  - Notify               |
|  - Redirect             |
+-------------------------+
```

8. JSCore Execution Architecture

```
+-------------------------+
|          JSCore         |
+-------------------------+
| - Send ApiRequest       |
| - Receive ApiResponse   |
| - Execute Commands      |
| - Update DOM            |
+-------------------------+
```

Command Execution Flow

```
ApiResponse JSON
       |
       v
JSCore Command Engine
       |
       v
DOM Manipulation
```

9. Translation Engine Architecture

```
+-------------------------+
|     Translation Engine  |
+-------------------------+
| - Detect Browser Language |
| - Lookup Dictionary       |
| - Replace Tokens          |
+-------------------------+
```

Translation Points

```
GET:
   HtmlDocument → Translate → Output

POST:
   ApiResponse → Translate → JSON
```

10. Security Architecture

```
+-------------------------+
|       Security Hooks    |
+-------------------------+
| OnInitialized()         |
| OnRequest()             |
| OnBeforeRender()        |
+-------------------------+
```

Security Flow

```
Request
   |
   v
Security Checks
   |
   v
Allow / Deny
```